(vi) Standard electrode potential (of an electrochemical reaction)

The *standard potential* of an electrochemical reaction, abbreviated as standard electrode potential, is defined as the standard potential of a hypothetical cell, in which the electrode (half-cell) at the left of the cell diagram is the *standard hydrogen electrode* (SHE) and the electrode at the right is the electrode in question. The standard hydrogen electrode consists of a platinum electrode in contact with a solution of H^+ at unit activity and saturated with H_2 gas with a fugacity referred to the standard pressure of 10^5 Pa (see Section 4.11.1 (v), p. 53). For a metallic electrode in equilibrium with solvated ions the cell diagram is

$$\mathrm{Pt} \mid \mathrm{H_2} \mid \mathrm{H^+} \,\vdots\, \mathrm{M^{z+}} \mid \mathrm{M}$$

and relates to the reaction

$$\mathrm{M^{z+}} + (z/2)\ \mathrm{H_2(g)} = \mathrm{M} + z\ \mathrm{H^+}$$

This diagram may be abbreviated $E(\mathrm{M^{z+}}/\mathrm{M})$. Note that the standard hydrogen electrode as defined is limited to aqueous solutions. See the Green Book for more information (Section 2.13.1 (vi), p. 74 in [20]).